Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/18/2016
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Assessing Telomere Lengths and Telomerase Activity in Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	10/24/2013	**End Date:**	06/23/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is 6/23/2018 per NSSC information and PI (Ed., 8/2/17) NOTE: End date changed to 10/23/2017 per NSSC information (Ed., 7/26/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive DNA sequence and associated proteins, which serve to protect chromosomal termini from inappropriate degradation. Functional telomeres also prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks; DSBs) and triggering inappropriate DNA damage responses, which could result in chromosomal instability. While it is well established that telomere length erodes with normal aging, it is becoming increasingly appreciated that telomere length is also influenced by a variety of other factors including life stress, infection, and inflammation. Inflammation also results in oxidative stress, both being known contributors to telomere erosion. Recent research supports telomere maintenance as a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and aging-related diseases. The rate of telomere length change, therefore, provides an informative biomarker of biological aging. Further, dysfunction/decline of telomere length can be linked to age-related pathologies, ranging from reduced immune function and loss of fertility, to cardiovascular disease and cancer. Preservation of telomere length requires telomerase, the specialized reverse transcriptase capable of de novo addition of telomeric repeats onto the very ends of newly synthesized telomeres. Telomerase activity is prominent in highly proliferative cells like stem, germline, and cancer, but virtually absent in normal somatic cells. Changes in diet and lifestyle have been shown to modulate telomerase activity, with healthy lifestyles and diets being positively correlated with telomere length. We've demonstrated that ionizing radiation also influences telomeres and telomerase. However, the relationships between telomere length and telomerase activity (measured in blood lymphocytes), lifestyle factors, and environmental exposures are largely unknown. We speculate that for the astronauts, telomere maintenance represents a particularly relevant biomarker, as it reflects the combined exposures and experiences encountered during space travel. That is, an individual astronaut's genetic susceptibilities, exposures to galactic space radiations, nutritional, physical, and psychological stressors encountered, are all captured as changes of telomere length maintenance; telomeres don't forget! Our overall goal is to identify and begin to define the risk of accelerated telomere shortening and changes in telomerase activity associated with spaceflight, a lifestyle with a variety of unique stressors. Recruitment of prospective astronauts (and age, gender-matched ground controls) is now complete, and monitoring of telomere lengths and telomerase activity is on-going; samples are being collected pre-flight (to establish baseline) and post-flight (to evaluate long-term/permanent changes). Together with future investigation of potential mechanisms (e.g., oxidative stress) and mitigation strategies (e.g., antioxidants), as well as data sharing related to e.g., diet and nutrition, stress and radiation exposure, results will help to establish relevant relationships and suggest potential interventions for further study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although relationships between telomere length and telomerase activity, and lifestyle factors (e.g., nutritional, physical, and psychological stressors) underlie aging and age-related degenerative pathologies (e.g., cardiovascular disease and cancer), they are largely unknown for human populations. Certainly aging and such age-related pathologies are an everyday concern on Earth as well as in space. The rate at which telomeres shorten provides an informative biomarker of biological aging that is influenced by a variety of lifestyle factors, including a variety of stressors and environmental exposures (e.g., UV and ionizing radiations). These on-going studies seek to make comparisons with and between unrelated astronauts (and age, gender-matched controls) that will serve to identify spaceflight and individual susceptibility factors that affect telomere length and telomerase activity. Results will begin to establish relevant relationships and suggest potential mitigation strategies for future study; therefore, they have the potential to improve overall health not only of astronauts, but of those on Earth as well.
Task Progress:	We finalized consenting and securing prospective astronaut (and age-matched ground test control) samples for this study, as our required minimum of 10 astronauts (and 10 corresponding test ground subjects) were acquired, at which time consent briefings were halted. Discussions continued, but we were still unable to obtain any retrospective (archived) samples as proposed for this project due to NASA sample sharing issues regarding the use of medical/research biospecimens; therefore, no retrospective samples have been analyzed. This is a most unfortunate situation, particularly in light of the fact that several in our prospective astronaut cohort have archived samples that could be used for telomere length analysis –tremendously valuable to the current study establishing rate of telomere loss associated with time in space. Inflight samples (R-1) and post-flight samples (R+7) were acquired from two Expedition 43S astronauts and their two ground test subject controls. Continuing baseline data collection included six (6) pre-flight samples from Expedition 45S, 46S, 47S, and 48S astronauts, and 6 from their respective ground test subjects. Post-flight samples (R+60) were acquired from two Expedition 43S astronauts and their two ground test subject controls. Continuing baseline data collection included six (6) pre-flight samples from Expedition 47S and 48S astronauts, and 6 from their respective ground test subjects. Final post-flight samples (R+180) were acquired from two Expedition 43S astronauts and their two ground test subject controls. Post flight samples (R+7) were acquired from Expedition 43S astronauts and their two ground test subject controls. Continuing baseline data collection included three (3) pre-flight samples from Expedition 48S and 49S astronauts, and 3 from their respective ground test subjects. All samples were processed and shipped successfully; data collection and analyses for telomere length measurement and telomerase activity assessment are on-going. Approaches and assays are continually undergoing optimization; e.g., we now have droplet digital (dd) PCR capabilities for quantitative telomerase activity determination with single cell resolution. Chromosome aberration analyses has been initiated on all samples and is also on-going. Expedition 45S R+60, 46S R+7, 49S L-60 collections are planned within the next three months (astronauts and test ground controls). Data collection and analyses for telomere length measurement, telomerase activity assessment, and chromosome aberration frequencies will proceed as per previous. As data collection for individual astronauts is completed (L-270, L-180, L-60; R+1-7, R+60, R+180, R+270), results are being complied and statistical analyses begun. We note that completion of sample collection for current cohort of astronauts in the study is projected well into 2018 (R+270 samples for latest recruits).

Bibliography Type:	Description: (Last Updated: 12/10/2025)
Abstracts for Journals and Proceedings	Bailey SM, McKenna MJ, Taylor L, George K. "Assessing Telomere length and Telomerase activity in Twin and Unrelated Astronauts." Invited presentation at 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	McKenna MJ, Taylor L, George K, Bailey SM. "Telomere Length and Telomerase Activity as Potential Biomarkers in Astronauts." Poster presentation at 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. , Sep-2015
Articles in Peer-reviewed Journals	Gao X, Sishc BJ, Nelson CB, Hahnfeldt P, Bailey SM, Hlatky L. "Radiation-induced reprogramming of pre-senescent mammary epithelial cells enriches putative CD44(+)/CD24(-/low) stem cell phenotype." Front Oncol. 2016 Jun 14;6:138. eCollection 2016. http://dx.doi.org/10.3389/fonc.2016.00138 ; PubMed PMID: 27379202; PubMed Central PMCID: PMC4905979 , Jun-2016
Articles in Peer-reviewed Journals	Sishc BJ, Nelson CB, McKenna MJ, Battaglia CL, Herndon A, Idate R, Liber HL, Bailey SM. "Telomeres and telomerase in the radiation response: Implications for instability, reprograming, and carcinogenesis." Front Oncol. 2015 Nov 24;5:257. eCollection 2015. http://dx.doi.org/10.3389/fonc.2015.00257 ; PubMed PMID: 26636039; PubMed Central PMCID: PMC4656829 , Nov-2015